James D. East, Ph.D.

Updated November 2026

Email: james@carbonmapper.org Website: https://eastjames.github.io

Education

Ph.D., Environmental Engineering 2023
North Carolina State University, Raleigh, NC
Faculty Advisor: Fernando Garcia Menendez

B.S., Environmental Engineering 2015
North Carolina State University, Raleigh, NC

Experience

Research Scientist Carbon Mapper, Inc.
10/2026 - present Boston, MA
Inverse modeling, remote sensing, atmospheric science.

Research Associate Harvard University
2/2026 - 10/2026 Cambridge, MA
School of Engineering and Applied Sciences
Faculty Mentor: Daniel Jacob

Postdoctoral Fellow Harvard University
2/2023 - 2/2026 Cambridge, MA
School of Engineering and Applied Sciences
Faculty Mentor: Daniel Jacob

Methane Remote Sensing Specialist U.N. Environment Programme
10/2025 - 10/2026
Research affiliate at the U.N. International Methane Emissions Observatory

ORISE Research Fellow U.S. Environmental Protection Agency
2020 - 2023 Research Triangle Park, NC
Office of Research and Development
Mentors: Barron H. Henderson, Sergey Napelenok

Graduate Research Assistant NC State University
2017 - 2020 Raleigh, NC
Department of Civil, Construction, and Environmental Engineering
Faculty Advisor: Fernando Garcia Menendez

Publications

n=7 as first author
∗Submitted
Google Scholar
ORCiD

∗**29** D. Cusworth, D. Varon, D. Jervis, J. McKeever, A. Turner, C. Frankenberg, J. Worden, A. Thorpe, J. Fahlen, S. Pandey, J. Maasakkers, I. Aben, A. Eldering, R. Duren, B. Conrad, T. Scarpelli, D. Bon, M. Huffman, I. Irakulis-Loitxate, L. Guanter, J. Schroeder, B. Hmiel, E. Sherwin, S. Crowell, E. Kort, A. Brandt, F. Reuland, M. Sargent, **East, JD**, and D. Jacob. Satellite remote sensing of methane point sources: A review of capabilities and applications. *EGUsphere*, 2026. https://doi.org/10.5194/egusphere-2026-6060

∗**28** **JD East**, DJ Jacob, HM Kamdar, J Bruno, LA Estrada, I Irakulis-Loitxate, MM San Martin, JE Johnson, G Mateo Garcia, D Jervis, DJ Varon, V Mancoridis, DH Cusworth, EK Reidy, and Y Chen. Annual mean methane emissions from individual oil-gas point sources worldwide inferred from multi-satellite plume-imaging observations. *EarthArXiv*, 2026. https://doi.org/10.31223/X5HF8K

∗**27** J. B. H. Bushey, M. Bertolacci, J. Benmergui, J. Kostinek, M. Sargent, M. Knapp, E. Kyzivat, S. Ayvazov, M. Russi, Z. Zhang, A. Chulakadabba, J. Chen, C. Chan Miller, S. Roche, L. A. Estrada, **JD East**, L. Yin, J. P. Williams, A. Himmelberger, R. Gautam, M. Omara, and S. C. Wofsy. Quantifying total methane emissions in the permian basin using mfim: the methanesat flexpart inverse model. *EGUsphere*, 2026. https://doi.org/10.5194/egusphere-2026-4693

∗**26** B Conrad, A Ayasse, D Bon, T Scarpelli, **JD East**, R Duren, and D Cusworth. Oil

and gas methane super-emitters drive emission intensities in several countries and basins globally. *Under review*, 2026

25 Z Chen, DJ Jacob, H Lin, N Balasus, SE Hancok, LA Estrada, **JD East**, Y Zhang, X Wang, M He, M Liu, and DJ Varon. Tropical wetland methane emissions and trends (2004–2023) inferred from landsat-based inundated vegetation data. *Environ. Sci. & Tech.*, 2026. https://doi.org/10.1021/acs.est.6c05412

24 J Worden, S Pandey, H Nesser, K Bowman, C Harkins, C Lyu, JD Maasakkers, D Gordon, DJ Jacob, LA Estrada, DJ Varon, **JD East**, L Schmeisser, and Z Qu. Top-Down Benchmark of U.S. Methane Inventories Reveals Regional Discrepancies in Activity-Based Estimates. *Atmos. Chem. Phys.*, 2026. https://doi.org/10.5194/acp-26-8855-2026

23 N Balasus, DJ Jacob, AA Bloom, **JD East**, LA Estrada, SE Hancock, M He, TA Mooring, AJ Turner, and JR Worden. 2019–2024 trends in African livestock and wetland emissions as contributors to the global methane rise. *Atmos. Chem. Phys.*, 2026. https://doi.org/10.5194/acp-26-4601-2026

22 M He, DJ Jacob, LA Estrada, DJ Varon, M Sulprizio, N Balasus, **JD East**, E Penn, DC Pendergrass, Z Chen, TA Mooring, JD Maasakkers, PG Brodrick, C Frankenberg, KW Bowman, and L Bruhwiler. Attributing 2019-2024 methane growth using tropomi satellite observations. *Science Advances*, 2026. https://doi.org/10.1126/sciadv.adz9007

21 X Wang, DJ Jacob, H Nesser, N Balasus, L Estrada, MP Sulprizio, DH Cusworth, TR Scarpelli, Z Chen, **JD East**, and DJ Varon. Quantifying urban and landfill methane emissions in the united states using tropomi satellite data. *Science Advances*, 2026. https://doi.org/10.1126/sciadv.adz9308

20 M Sparks, **JD East**, F Garcia-Menendez, E Monier, and R Saari. Air quality alerts, health impacts, and adaptation implications under varying climate policy. *Environ. Sci. & Tech.*, 2026. https://doi.org/10.1021/acs.est.5c12522

19 LA Estrada, DJ Jacob, M He, **JD East**, DJ Varon, N Balasus, SE Hancock, M Sulprizio, KW Bowman, JR Worden, E Reidy, and BRK Runkle. Quantifying national, state, and oil/gas field methane emissions and trends in the U.S. (2019–2024) through high resolution inversion of satellite observations. *Atmos. Chem. Phys.*, 2026. https://doi.org/10.5194/acp-26-10629-2026

18 **JD East**, DJ Jacob, D Jervis, N Balasus, LA Estrada, SE Hancock, MP Sulprizio, J Thomas, X Wang, Z Chen, DJ Varon, and J Worden. Worldwide inference of national methane emissions by inversion of satellite observations with UNFCCC prior estimates. *Nature Communications*. https://doi.org/10.1038/s41467-025-67122-8

17 DJ Varon, JD Jacob, LA Estrada, N Balasus, **JD East**, DC Pendergrass, Z Chen, MP Sulprizio, M Omara, R Gautam, ZR Barkley, FJ Cardoso-Saldaña, EK Reidy, H Kamdar, ED Sherwin, S Biraud, D Jervis, S Pandey, J Worden, KW Bowman, JD Maasakkers, and RL Kleinberg. Seasonality and declining intensity of methane emissions from the permian and nearby us oil and gas basins. *Environ. Sci. & Tech.*, 2025. https://doi.org/10.1021/acs.est.5c08745

16 CA Quinn, T Colligan, EJ Ward, **JD East**, Y Lim, E Lee, R Koster, and B Poulter. Amazonia Wetland Methane Emission Decrease in 2023: Seasonal Forecasting of Global Wetlands Highlights Monitoring Targets in Critical Ecosystems. *JAMES*, 2025. https://doi.org/10.1029/2025MS005510

15 SE Hancock, DJ Jacob, R Jimenez, A Ardila, L Morales-Rincon, N Rojas, LA Estrada, N Balasus, **JD East**, MP Sulprizio, X Wang, JL France, L Potyk, E Penn, Z Chen, DJ Varon, C Frankenberg, M Baranski, A Calcan, and RJ Parker. Applying satellite observations to improve bottom-up national emission inventories for methane: Application to Colombia. *Atmos. Chem. Phys.*, 2026. https://doi.org/10.5194/acp-26-10455-2026

14 DC Pendergrass, DJ Jacob, N Balasus, L Estrada, DJ Varon, **JD East**, M He, TA Mooring, E Penn, H Nesser, and JR Worden. Trends and seasonality of 2019–2023 global methane emissions inferred from a localized ensemble transform Kalman filter (CHEEREIO v1.3.1) applied to TROPOMI satellite observations. *Atmos. Chem. Phys.*, 2025. https://doi.org/10.5194/acp-25-14353-2025

13 LA Estrada, DJ Varon, MP Sulprizio, H Nesser, Z Chen, N Balasus, SE Hancock, M He, **JD East**, TA Mooring, AO Alonso, JD Maasakkers, I Aben, S Baray, KW Bowman, JR Worden, FJ Cardoso-Saldaña, E Reidy, and DJ Jacob. Integrated Methane Inversion (IMI) 2.0: an improved research and stakeholder tool for monitoring total methane emissions with high resolution worldwide using TROPOMI satellite observations. *Geosci. Model Dev.*, 2025. https://doi.org/10.5194/gmd-18-3311-2025

12 Z Chen, JD Jacob, H Lin, N Balasus, A Hardy, **JD East**, Y Zhang, BR Runkle, SE Hancock, CA Taylor, X Du, and BO Sander. Global Rice Paddy Inventory (GRPI): a high-resolution inventory of methane emissions from rice agriculture based on Landsat satellite inundation datat. *Earth's Future*, 2025. https://doi.org/10.1029/2024EF005479

11 E Penn, DJ Jacob, Z Chen, **JD East**, MP Sulprizio, L Bruhwiler, JD Maasakkers, H Nesser, Z Qu, Y Zhang, and W Worden. What can we learn about tropospheric OH from satellite observations of methane? *Atmos. Chem. Phys.*, 2025. https://doi.org/10.5194/acp-25-2947-2025

10 LH Yang, DJ Jacob, H Lin, R Dang, KH Bates, **JD East**, KR Travis, DC Pendergrass, and LT Murray. Model underestimates of OH reactivity cause overestimate of hydrogen's climate impact. *Geophysical Research Letters*, 52:e2024GL112445, 2025. https://doi.org/10.1029/2024GL112445

9 Q Zhu, DJ Jacob, K Yuan, F Li, BRK Runkle, M Chen, AA Bloom, B Poulter, **JD East**, WJ Riley, G McNicol, J Worden, C Frankenberg, and M Halabisky. Advancements and opportunities to improve bottom-up estimates of global wetland methane emissions. *Environmental Research Letters*, 20:023001, 2025. https://doi.org/10.1088/1748-9326/adad02

8 SE Hancock, DJ Jacob, Z Chen, H Nesser, A Davitt, DJ Varon, MP Sulprizio, N Balasus, LA Estrada, M Cazorla, L Dawidowski, S Diez, **JD East**, E Penn, CA. Randles, JR Worden, I Aben, RJ Parker, and JD Maasakkers. Satellite quantification of methane emissions from South American countries: A high-resolution inversion of TROPOMI and GOSAT observations. *Atmospheric Chemistry and Physics*, 25:797–817, 2025. https://doi.org/10.5194/acp-25-797-2025

7 **JD East**, E Monier, RK Saari, and F Garcia-Menendez. Projecting changes in the frequency and magnitude of ozone pollution events under uncertain climate sensitivity. *Earth's Future*, 12(6):e2023EF003941, 2024. https://doi.org/10.1029/2023EF003941

6 **JD East**, DJ Jacob, N Balasus, AA Bloom, LP Bruhwiler, Z Chen, JO Kaplan, LJ Mickley, TA Mooring, E Penn, B Poulter, MP Sulprizio, JR Worden, RM Yantosca, and Z Zhang. Interpreting the seasonality of atmospheric methane. *Geophysical Research Letters*, 51(10), 2024. https://doi.org/10.1029/2024GL108494

**Ranked within the top 10% of most-viewed papers published by the journal in 2024*

5 Matt S. Sparks, I Farahbakhsh, M Anand, CT Bauch, KC Conlon, **JD East**, T Li, M Lickley, F Garcia-Menendez, E Monier, and RK Saari. Health and equity implications of individual adaptation to air pollution in a changing climate. *Proceedings of the National Academy of Sciences*, 121(5):e2215685121, 2024. https://doi.org/10.1073/pnas.2215685121

4 **JD East**, E Monier, and F Garcia-Menendez. Characterizing and quantifying uncertainty in projections of climate change impacts on air quality. *Environmental Research Letters*, 17(9), 2022. https://doi.org/10.1088/1748-9326/ac8d17

3 **JD East**, BH Henderson, SL Napelenok, SN Koplitz, G Sarwar, R Gilliam, A Lenzen, DQ Tong, RB Pierce, and F Garcia-Menendez. Inferring and evaluating satellite-based constraints on NO_x emissions estimates in air quality simulations. *Atmospheric Chemistry and Physics*, 22(24):15981–16001, 2022. https://acp.copernicus.org/articles/22/15981/2022/

2 Daiwen Kang, C Hogrefe, S Golam, **JD East**, JM Madden, R Mathur, and BH Henderson. Assessing the Impact of Lightning NOx Emissions in CMAQ Using Lightning Flash Data from WWLLN over the Contiguous United States. *Atmosphere*, 13(8), 2022. https://www.mdpi.com/2073-4433/13/8/1248

1 **JD East**, JS Montealegre, JE Pachon, and F Garcia-Menendez. Air quality modeling to inform pollution mitigation strategies in a Latin American megacity. *Science of The Total Environment*, 776(145894), 2021. https://doi.org/10.1016/j.scitotenv.2021.145894

Additional Publications

- **JD East** and F Garcia-Menendez. *Internal climate variability and initial condition ensembles in air quality projections.* In: C. Deser and K. Rodgers (guest eds) New research on climate variability and change using initial-condition Large Ensembles. Special issue of US CLIVAR VARIATIONS. Volume 18, Number 2, Summer 2020. http://dx.doi.org/10.5065/0DSY-WH17

- Fernando Garcia-Menendez, **JD East**, BD Pienkosz, and E Monier. Climate model response uncertainty in projections of climate change impacts on air quality. In Wanmin G Mensink C and Hakami A, editors, *Air Pollution Modeling and its Application XXVI*, pages 433–437. Springer International Publishing, 2020. https://doi.org/10.1007/978-3-030-22055-6_69

- Juan S. Montealegre, J Vanegas, JE Pachon, A Rojas, **JD East**, and F Garcia-Menendez. Air quality modeling as a tool for adjusting emission inventories. In *2019 Congreso Colombiano y Conferencia Internacional de Calidad de Aire y Salud Pública (CASAP)*, pages 1–4, 2019. https://doi.org/10.1109/CASAP48673.2019.9364063

Invited Talks

2026 North Carolina State University. Raleigh, NC. September 2026.

2026 University of North Carolina. August 2026.

2026 Environmental Defense Fund, MethaneSAT Science Meeting. June 2026.

2026 Union College. Schenectady, NY. February 2026.

2025 Michigan State University. East Lansing, MI. March 2025.

2024 Air & Waste Management Association - RTP Chapter. April 2024.

2024 Harvard University SEAS Cross Area Seminar. Cambridge, MA. March 2024.

2024 NASA Jet Propulsion Laboratory. Pasadena, CA. January 2024.

2023 AGU Atmospheric and Space Electricity Early Career seminar. June 2023.

2023 NASA Atmospheric Chemistry and Dynamics Lab Seminar. January 2023.

2022 HAQAST Update22 speaker and panelist (link). January 2022.

Selected Conference Presentations

2026 22nd International Workshop on Greenhouse Gas Measurements from Space (IWGGMS-22). Bonn, Germany. July 1, 2026.

2026 12th International GEOS-Chem Meeting. St. Louis, MO. June 9, 2024.

2025 EGU General Assembly 2025, Vienna, Austria. May 1, 2025. EGU25-14476

2024 AGU24, Washington, DC. December 12, 2024 B43K-03

2024 11th International GEOS-Chem Meeting. St. Louis, MO. June 13, 2024.

2023 Task Force on Hemispheric Transport of Air Pollution. April 21, 2023.

2022 AGU Fall Meeting, Chicago, IL. December 13, 2022. AE13A-08

2022 CMAS Conference, Chapel Hill, NC. October 18, 2022.

2022 Environmental Engineering Symposium at NC State. March 4, 2022.

2022 HAQAST Update22. January 20, 2022.

2021 CMAS Conference, Chapel Hill, NC. November 2, 2021.

2021 Science on Earth Day "ScED" Talks, U.S. EPA, ORD. April 22, 2021.

2021 Early Career Seminar. NLTO. U.S. EPA. March 25, 2021.

2020 CMAS Conference, Chapel Hill, NC. October 27, 2020.

2019 CASAP Conference. Barranquilla, Colombia. August 14, 2019.

2019 NC Breathe Conference. Wilmington, NC. April 11, 2019.

Selected Poster Presentations

2026 Methane250, Ispra, Italy. March 29, 2026.

2025 AGU Fall Meeting, New Orleans, LA. December 19, 2025. A51T-1033

2025 NASA Carbon Monitoring System science team meeting. September 9, 2025.

2023 AGU Fall Meeting, San Francisco, CA. December 12, 2023. B21K-2111

2022 AGU Fall Meeting, Chicago, IL. December 16, 2022. A52N-1166

2022 11th International GEOS-Chem Conference. June 8, 2022.

2022 TEMPO Science Team Meeting. June 1, 2022.

2021 AGU Fall Meeting, New Orleans, LA. December 13, 2021. A15B-1625

2021 Environmental Engineering Symposium at NC State. February 26, 2021.

2020 AGU Fall Meeting. December 10, 2020. GC060-0006

2020 AAAR Conference. October 5, 2020.

2020 HAQAST Showcase. July 21, 2020.

2019 NC State University Latin American Research Symposium. February 2019.

2018 111th Air & Waste Management Association, Hartford, CT. June 25, 2018.

2015 108th Air & Waste Management Association, Raleigh, NC. June 22, 2015.

Teaching & Mentoring

Teaching Assistant

Fundamentals of Environmental Engineering. NC State University. Spring 2018.

- Weekly office hours. Graded tests and assignments. Solutions for class site.

Guest Lecturer

Environmental Modeling. NC State University. Spring 2022.

- Lecture and active learning jupyter notebook activity.

Research Mentoring

Gabrielle Diroll, undergraduate, UC Berkeley

- 2026 Harvard summer research program (SPHEER)

Sophie Farr, undergraduate, Vassar College

- 2024 Harvard summer research program (SPHEER). AGU24 poster.

Grace Gould, undergraduate, NC State University

- 2021 NC State University REU Summer Program. Research symposium poster.

Funding & Awards

Grants and Fellowships

(declined) NASA Postdoctoral Program, Jet Propulsion Laboratory. 2023. $92,187

U.S. Environmental Protection Agency ORISE Fellowship. 2020-2023. $180,000

DJ Jacob and **JD East** *(co-I)*. "Integration of gridded GHGSat and TROPOMI satellite data to quantify methane emissions on national and regional scales." GHGSat, Inc. May 2024 - April 2026. $100,000.

JD East and DJ Jacob. "Construction of Jacobian matrices for inversion of satellite methane observations." Harvard FAS Research Computing. January 2024. 870,912 core-hours.

Awards

Finalist. Three Minute Thesis. University level. NC State University. 2022.
Talk video | Dept. News Article

1st Place. Three Minute Thesis. Department level. NC State University. 2022.
Dept. News Article

1st Place. Masters student poster competition. AWMA ACE. 2018.
Dept. News Article

Honorable Mention Poster. EWC Symposium, NC State University. 2018.

1st Place. Undergrad student poster competition, AWMA ACE. 2015.
Dept. News Article

Academic Recognition

Sustainability Research and Study Related to Air Quality and Waste Management Award. Air & Waste Management Association. 2019.

Graduate Merit Award. Dept. of Civil, Construction, and Environmental Engineering. NC State University. 2017.

Travel Funding

NC State Graduate School competitive funding award. Summer 2019.

Synergistic Activities

Science Team Member NASA Carbon Monitoring System. Methane Working Group. 2025 - present.

Session Convener AGU24, AGU25, AGU26. Atmospheric Sciences Section. Targeting Methane Mitigation: Quantification of Anthropogenic Methane Sources at All Scales Through Atmospheric Measurement (Session).

Co-lead Methane subgroup. Atmospheric Chemistry Modeling Group, Harvard University. 2024 - present.

Peer-review *Nature Communications; Environmental Science & Technology; JGR: Atmospheres; Biogeosciences; Atmospheric Chemistry & Physics; Scientific Data; Environmental Pollution*

Seminar Coordinator Atmospheric & Environmental Chemistry Seminar. Harvard University. Spring 2024.

Media engagement Marketplace Morning Report; Bloomberg Green; AAMC News

Participant EPA Methane Inverse Modeling Technical Workshop. Research Triangle Park, NC. July 2024.

Guest instructor Boy Scouts Merit Badge College. NC State University. Spring 2018. Taught about global air pollution.

A&WMA Student Chapter NC State University. Secretary, 2019-2020. President, 2018-2019.

Professional Memberships

American Geophysical Union, European Geophysical Union, A&WMA